

Constant-background unstable-mode counting does not bound large-field entropy

Later corrections: the action lower-bound note, §§2–4 and the transfer note, §3 distinguish action cost from normalized probability and field thresholds; the instability note, §§2–4 corrects the growth rate and withdraws sharp truncation claims (relevant passages read).

Correction (2026-10-02, GPT-6 Astra). The former claims that the entropy question is settled, that the large-field sum converges for $g^2 < 2\pi^2$, and that the expansion is merely slow are **withdrawn**. The assumed bounded factor per unstable mode was not proved and is not a count of regions or polymers. Compactness proves existence of a finite-lattice constrained minimizer, not a controlled Gaussian expansion; the assertion that its Hessian is nonnegative on the whole tangent cone was false. The assertion that the minimizer must be inhomogeneous is also withdrawn: a saddle direction must be feasible for the particular constraint. The count $\eta^2/(8\pi^2)$ survives only as a phase-space estimate for one unit-charge complex field, with $\eta = gB\ell^2$, not an exact finite-box count. Real directions double that estimate and $SU(3)$ has two additional charged pairs. The old ratio $2\pi^2/g^2$ divided an assumed cost by this model count; it is not a general action/entropy theorem. Section 3 fixes the field-norm and multiplicity conventions explicitly. The old net exponent is retained as conditional bookkeeping only.

Abstract / result. For a constant abelian chromomagnetic background, write $b = gB > 0$ for the curvature seen by a unit-charge component and $\eta_b = b\ell^2$. One complex charged field has unstable-mode phase-space count

$$\mathcal{N}_C^{\text{ps}} = \frac{b\ell^2}{2\pi} \frac{b\ell^2}{4\pi} = \frac{\eta_b^2}{8\pi^2}.$$

This counts the lowest aligned Landau branch with $k_3^2 + k_4^2 < b$. It has no explicit block-size factor at fixed η_b , but is a continuum density approximation, with boundary, flux and momentum discreteness qualifications. It is not a uniform lattice estimate or a bound for inhomogeneous backgrounds. The continuum flowed-block action floor $\eta_F^2/(4g^2)$ is a separate result with its own L^2 threshold η_F and flow assumptions. Neither ingredient proves normalized large-field rarity, an entropy bound, polymer convergence or a continuum mass gap.

1. Compactness gives a minimizer, not a Gaussian expansion

Proposition 1 (finite regulator only). Let G be compact, let $\mathcal{E}$ be a finite link set, and let $K \subset G^\mathcal{E}$ be closed and nonempty. Every continuous action s attains its minimum on K . If s is real valued, its integral $\int_K e^{-s} d\nu$ is finite for every finite reference measure ν .

Proof. The product and its closed subset are compact. The extreme value theorem gives a minimum and a finite maximum; e^{-s} is bounded, so its integral against a finite measure is finite. $\square$

This says nothing about regulator-uniform estimates, the size of a Gaussian contribution or convergence of a saddle-point expansion. For a C^2 action, a constrained minimum requires nonnegative first variation along one-sided feasible curves. At a boundary minimum the first variation need not vanish, so the Hessian need not be nonnegative on the tangent cone. In a local coordinate, take

$$K = [0, 1/4], \quad s(x) = x - x^2.$$

The minimum is at 0, with $s'(0) = 1$ but $s''(0) = -2$, even though positive directions belong to the tangent cone. Along a twice differentiable feasible curve γ whose first action derivative

vanishes, the necessary condition is

$$(s \circ \gamma)''(0) = \text{Hess } s[\dot{\gamma}, \dot{\gamma}] + ds[\ddot{\gamma}] \geq 0$$

in local coordinates. Constraint curvature can matter as well. At an interior minimum $ds = 0$ and the Hessian is nonnegative, but it can have zero modes: $s(x) = x^4$ has no coercive quadratic term at its minimum. Gauge degeneracy, boundary geometry, higher terms and uniform determinant bounds need separate treatment.

The corrected instability note, §2 (passage read) supplies a negative quadratic mode for a stationary constant background in its specified background-transverse sector. It excludes that background as a constrained minimum only if an admissible perturbation realizes the descent while remaining in K . No such feasibility proof, or construction of an inhomogeneous minimizer, is supplied here.

2. Counting the unstable directions in a specified background

Use the continuum background-covariant operator of the instability note, §2 (passage read), in four Euclidean dimensions. For one charged complex field, with charge magnitude $q > 0$ relative to T^3 , its transverse polarizations have eigenvalues

$$\omega^2 = k_3^2 + k_4^2 + (2n + 1)qb \mp 2qb.$$

Only $n = 0$ with aligned spin is negative, for $k_3^2 + k_4^2 < qb$. Neutral components have no such branch.

Landau degeneracy. The transverse density per unit area is $qb/(2\pi)$. For example, in Landau gauge with transverse length ℓ and periodic second coordinate, momenta are spaced by $2\pi/\ell$. Guiding centres are therefore spaced by $2\pi/(qbl)$, giving that density. On a magnetic torus with compatible flux, $qbl^2/(2\pi)$ is an integer degeneracy. For a block cut from the plane it is a bulk density, not an exact boundary-condition-independent count.

Longitudinal count. Replacing a momentum sum by its phase-space integral gives

$$\left(\frac{\ell}{2\pi}\right)^2 \text{area}\{k_3^2 + k_4^2 < qb\} = \frac{qb\ell^2}{4\pi}.$$

Thus the **phase-space estimate**, in complex dimension, is

$$\mathcal{N}_{\mathbb{C}}^{\text{ps}}(q) = \frac{q^2 b^2 \ell^4}{8\pi^2} = \frac{q^2 \eta_b^2}{8\pi^2}.$$

An exact periodic longitudinal count, when the magnetic flux and boundary conditions admit this background, would instead be

$$\mathcal{N}_{\mathbb{C}}^{\text{box}}(q) = m_q \#\{n \in \mathbb{Z}^2 : 4\pi^2 |n|^2 < q\eta_b\}, \quad m_q = \frac{q\eta_b}{2\pi} \in \mathbb{N}.$$

The disc area is an approximation to this integer count, justified as $q\eta_b \rightarrow \infty$: unit squares around the integer points place it between disc areas with radii differing by a bounded constant. It is not justified merely by sending $\ell/a \rightarrow \infty$ at fixed η_b . A lattice operator also has dispersion and cutoff effects; no uniform lattice replacement is proved here.

Real and colour multiplicities. A charged pair of real adjoint components forms one complex field. Each complex mode has two real amplitudes, so its real negative-space dimension is twice the complex count. Take the usual $SU(3)$ generator

$$T^3 = \text{diag}(1/2, -1/2, 0).$$

The matrix units satisfy $[T^3, E_{ij}] = (t_i - t_j)E_{ij}$, where t_i are its diagonal entries. The three conjugate root pairs therefore have charge magnitudes 1, 1/2, 1/2. Each pair is counted once as a complex field; adding its conjugate again would double-count it. Consequently

$$\begin{aligned}\mathcal{N}_{\mathbb{C}, SU(3)}^{\text{ps}} &= (1 + 1/4 + 1/4) \frac{\eta_b^2}{8\pi^2} = \frac{3\eta_b^2}{16\pi^2}, \\ \mathcal{N}_{\mathbb{R}, SU(3)}^{\text{ps}} &= \frac{3\eta_b^2}{8\pi^2}.\end{aligned}$$

These are estimates for this background and operator sector, not a mode count for every large-field configuration or a gauge-fixed functional determinant calculation.

3. Action cost against model count: fix the normalization first

The corrected transfer note, §3 and action note, §2 (passages read) use

$$s(A) = \frac{S_E[A]}{\hbar} = \frac{1}{4g^2} \int_{\Omega} |F|^2 d^4x,$$

where the norm sums colour and **ordered** spacetime indices. For a smooth finite-action field admitting action-decreasing flow through $t = \ell^2/8$, with no boundary flux in the dissipation identity, the constraint $\int_B |G_t|^2 \geq \eta_F^2$ gives

$$s(A) \geq s(B_t) \geq \frac{1}{4g^2} \int_B |G_t|^2 \geq \frac{\eta_F^2}{4g^2}.$$

This is a classical continuum action inequality, not a normalized measure bound. Its equality or sharpness is not established by a constant field supported only on a block.

The spectral parameter $\eta_b = b\ell^2$ of Section 2 is not implicitly η_F . To compare them in the homogeneous model, fix the colour norm by $\langle T^3, T^3 \rangle = 1$ (equivalently $2\text{tr}[(T^3)^2] = 1$ in the Hermitian convention), and absorb the coupling into the curvature. With $F_{12} = bT^3$, $F_{21} = -bT^3$ and all other components zero,

$$|F|^2 = 2b^2, \quad \eta_F^2 = 2\eta_b^2, \quad s_B := \frac{1}{4g^2} \int_B |F|^2 = \frac{\eta_b^2}{2g^2}.$$

Here s_B is the block action of the background, not the minimum action for an arbitrary constrained field. No finite-action localization or admissible boundary construction follows from this calculation.

Proposition 2 (model ratios, not entropy bounds). For $\eta_b > 0$ and the preceding normalization, division by the phase-space counts of Section 2 gives the following identities:

count convention	phase-space count	$s_B/\mathcal{N}^{\text{ps}}$
one unit-charge complex field	$\eta_b^2/(8\pi^2)$	$4\pi^2/g^2$
its real directions	$\eta_b^2/(4\pi^2)$	$2\pi^2/g^2$
all three $SU(3)$ charged pairs, complex	$3\eta_b^2/(16\pi^2)$	$8\pi^2/(3g^2)$
all three $SU(3)$ charged pairs, real	$3\eta_b^2/(8\pi^2)$	$4\pi^2/(3g^2)$

Proof. Substitute the counts and $s_B = \eta_b^2/(2g^2)$, and cancel η_b^2 . $\square$

The original arithmetic

$$\frac{\eta_b^2/(4g^2)}{\eta_b^2/(8\pi^2)} = \frac{2\pi^2}{g^2}$$

remains a formal identity if the numerator is an **assumed** cost s_{old} . It paired the ordered-index L^2 threshold formula with the different spectral threshold without their conversion. Under the normalization above, the same number occurs for one pair's real directions, not its complex count. None of these ratios is a universal entropy coefficient. Scale cancellation at fixed η_b and g does not establish cutoff uniformity or running-coupling control.

The missing weight estimate. For a finite regulator with reference probability measure ν and $d\mathbb{P} = Z^{-1}e^{-s}d\nu$, an action floor $s \geq Q$ on an event K gives only

$$\mathbb{P}(K) \leq Z^{-1}e^{-Q}\nu(K).$$

The denominator needs control relative to the constrained numerator. The negative eigenspace dimension at one saddle controls neither this ratio nor the volume of a region, its nonlinear integration, or the number of possible regions. Compactness at a fixed regulator provides no uniform factor per mode.

To preserve the former calculation's scope explicitly, **if one assumed** a weight bound

$$W(\eta_b) \leq e^{-s_{\text{old}}}e^{C\mathcal{N}_{\text{c}}^{\text{ps}}(1)}, \quad s_{\text{old}} = \frac{\eta_b^2}{4g^2}, \quad C > 0,$$

then algebra would give

$$W(\eta_b) \leq \exp \left[-\frac{\eta_b^2}{4g^2} \left(1 - \frac{Cg^2}{2\pi^2} \right) \right].$$

Its exponent is negative for $g^2 < 2\pi^2/C$. Neither the assumed weight estimate nor a regulator-independent C is proved. The old numerical threshold $g^2 < 2\pi^2$ additionally set $C = 1$ without justification. This display is **conditional bookkeeping**, not a convergence theorem. Even an actual tail in strength would need region-counting and interaction estimates to sum polymers.

4. What survives and what remains open

Survives. Finite-lattice compactness, the constant-background spectrum, and the phase-space counts with the stated boundary, charge and real multiplicity conventions. The separate continuum action floor survives under its flow and boundary assumptions. Ratios in Proposition 2 compare two background model quantities.

Withdrawn. The inference that suppression beats large-field entropy, the convergence threshold, the claim that the expansion is merely slow, and the assertion that the remaining task is only to identify an inhomogeneous minimizer. The saddle calculation does not determine the geometry of a constrained minimizer or its Gaussian order.

Open estimates. A normalized local large-field weight bound with volume, cutoff and boundary dependence; control of inhomogeneous fields and their integration; summability over locations, shapes and interacting polymers; and, if a constrained expansion is used, feasibility, gauge treatment and control of its nonquadratic terms. The former proposed weight $\exp[-\eta_b^2(1 - O(g^2))/(4g^2)]$ is at most a heuristic target under the old cost convention, not a prediction of Proposition 2. A measure estimate would still not supply the signed Hamiltonian comparison discussed in the operator-inequality note, §3 (passage read).

5. Consequence for STATE

Retain the constant-background phase-space model and explicitly normalized ratios, alongside the separate continuum action floor. Large-field entropy control, normalized rarity, a controlled Gaussian expansion and polymer convergence remain open here; the earlier closure and weak-coupling convergence claims are withdrawn. No continuum $SU(3)$ construction or mass gap follows. This section specifies the corrected scope for the caller's STATE and catalog updates; those files are unchanged in this task.